

All- k Two-Radical Kummer Closures

A canonical norm-derived radicand and maximal wreath monodromy

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Abstract

For the generic multiquadratic mass cover

$$w_i^2 = u + N_i^2, \quad 4M = \sum_{i=1}^k w_i,$$

the base mass field has degree

$$\delta_k = \begin{cases} 2^{k-1}, & k \text{ odd}, \\ 2^{k-1} - \frac{1}{2} \binom{k}{k/2}, & k \text{ even}, \end{cases}$$

and generic normal-closure group S_{δ_k} . We construct, for every $k \geq 3$, a coefficient-free second radicand

$$\gamma_k = 2 \left(\sum_{r=0}^{\lfloor k/2 \rfloor} u^r e_{k-2r}(w) + \prod_{i=1}^k N_i \right)$$

that specializes at $k = 4$ to the static horizon radicand. Its companion polynomial β_k satisfies the uniform identity

$$\gamma_k(\gamma_k - 4P) = 4u\beta_k^2, \quad P = \prod_i N_i.$$

Even and odd signed-contact divisors give valuation rows $(1, 0)$ and $(1, 1)$ for the channels (u, γ_k) . We prove that all off-sheet entries are even, so their Galois orbits form $B \otimes I_{\delta_k}$ with $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Consequently all $2\delta_k$ conjugate square classes are independent. The normal closure of $K(\sqrt{u}, \sqrt{\gamma_k})/F$ therefore has the exact group

$$C_2^2 \text{ wr } S_{\delta_k}$$

and order $2^{2\delta_k} \delta_k!$. This proves the all- k static two-channel Kummer closure. The theorem is algebraic for arbitrary k ; only the four-charge member is asserted here to carry the established black-hole entropy interpretation.

1 Introduction

The four-charge horizon cover carries two quadratic channels above its quintic mass field: the axial class $[u]$ and an entropy-sum class $[\gamma_4]$. Signed contact divisors separate these channels and yield the static closure group $C_2^2 \text{ wr } S_5$ [4, 3]. It was not previously clear whether the second class was forced by the general multiquadratic norm structure or was an accident of the four-charge truncation.

The issue is subtle only if “second radicand” is constrained. An arbitrary function field has many independent square classes, so searching for some additional class would say nothing

about the horizon construction. The candidate used here is instead forced by the two axial products

$$A_k^+ = \prod_i (w_i + m), \quad A_k^- = \prod_i (w_i - m), \quad m^2 = u.$$

Their product is the square P^2 , where $P = \prod_i N_i$. Thus the sum-compatible expression

$$A_k^+ + A_k^- + 2P$$

is the square of a sum after adjoining compatible square roots of A_k^+ and A_k^- . It is invariant under $m \mapsto -m$, uses no adjustable coefficients or selected auxiliary divisor, and recovers the published four-charge normalization. These properties pin the relevant algebraic generalization.

Exact $k = 3$ and $k = 6$ falsifier computations first exposed the uniform identity used below [5]. The proof itself does not interpolate from those cases. It works for arbitrary $k \geq 3$, proves the required local valuation statements directly, and then invokes the already established all- k symmetric base monodromy [1] and generic reflection descent [2].

Scope. The result is a theorem about the generic equal-weight multiquadratic mass cover and its canonical norm-derived decoration. At $k = 4$ the decoration is the physical static entropy-sum channel. For other values of k , the theorem asserts the algebraic closure of the displayed cover; it does not invent a black-hole solution with an arbitrary number of independent charges.

2 The generic mass cover

Fix an integer $k \geq 3$. Let

$$F_0 = \mathbb{Q}(N_1, \dots, N_k), \quad F = F_0(M),$$

where the N_i and M are algebraically independent. On the multiquadratic curve over F_0 put

$$w_i^2 = u + N_i^2 \quad (1 \leq i \leq k), \quad \sigma = \sum_{i=1}^k w_i. \quad (1)$$

The mass fibre is $\sigma = 4M$. Let $\mathcal{N}_k(u; M)$ denote the irreducible mass polynomial, let u be one of its roots, and put

$$K = F(u).$$

We use two established results. They are stated here in precisely the form needed by the new argument.

Theorem 2.1 (All- k base monodromy [1]). *The extension K/F is generically separable of degree*

$$\delta_k = \begin{cases} 2^{k-1}, & k \text{ odd}, \\ 2^{k-1} - \frac{1}{2} \binom{k}{k/2}, & k \text{ even}. \end{cases} \quad (2)$$

Its normal closure E/F has Galois group

$$\text{Gal}(E/F) \cong S_{\delta_k},$$

acting in the natural way on the δ_k embeddings of K into E .

Theorem 2.2 (Reflection descent [2]). *Let $m^2 = u$ and let W be the generic signed mass field containing $m, w_1, \dots, w_k$. The involution*

$$\tau(m) = -m, \quad \tau(w_i) = w_i$$

has fixed field K . In particular,

$$w_i \in K \quad (1 \leq i \leq k).$$

The descent statement is crucial: the symmetric expressions constructed below are not merely functions on an auxiliary radical sheet. They belong to the degree- δ_k mass field itself.

3 The canonical second radicand

Write $e_j(w) = e_j(w_1, \dots, w_k)$ for the j th elementary symmetric polynomial, with $e_0(w) = 1$, and put

$$\alpha_k = \sum_{r=0}^{\lfloor k/2 \rfloor} u^r e_{k-2r}(w), \quad (3)$$

$$\beta_k = \sum_{r=0}^{\lfloor (k-1)/2 \rfloor} u^r e_{k-1-2r}(w). \quad (4)$$

By Theorem 2.2, both elements lie in K .

Lemma 3.1 (Parity decomposition). *In $K(m)$ one has*

$$A_k^+ := \prod_{i=1}^k (w_i + m) = \alpha_k + m\beta_k, \quad A_k^- := \prod_{i=1}^k (w_i - m) = \alpha_k - m\beta_k. \quad (5)$$

Proof. Expanding the first product by elementary symmetric functions gives

$$\prod_{i=1}^k (w_i + m) = \sum_{j=0}^k m^{k-j} e_j(w).$$

The terms with $k - j$ even are exactly

$$u^{(k-j)/2} e_j(w), \quad j \equiv k \pmod{2},$$

and their sum is α_k . The remaining terms contain one factor of m ; after removing it, their sum is β_k . Replacing m by $-m$ fixes the even part and negates the odd part, proving both identities. $\square$

Proposition 3.2 (Uniform norm identity). *Let*

$$P = \prod_{i=1}^k N_i.$$

Then

$$\alpha_k^2 - u\beta_k^2 = P^2. \quad (6)$$

Consequently, for

$$\gamma_k = 2(\alpha_k + P), \quad (7)$$

one has

$$\boxed{\gamma_k(\gamma_k - 4P) = 4u\beta_k^2.} \quad (8)$$

Proof. Multiplying the two products in (5) gives

$$A_k^+ A_k^- = \prod_{i=1}^k (w_i^2 - m^2) = \prod_{i=1}^k N_i^2 = P^2,$$

because $w_i^2 - m^2 = (u + N_i^2) - u = N_i^2$. The left-hand side is $\alpha_k^2 - u\beta_k^2$, proving (6). Finally,

$$\gamma_k(\gamma_k - 4P) = 4(\alpha_k + P)(\alpha_k - P) = 4(\alpha_k^2 - P^2) = 4u\beta_k^2.$$

□

Proposition 3.3 (Canonicity and exhaustion of the cross-term choices). *Choose square roots $x_\pm^2 = A_k^\pm$ compatibly so that $x_+ x_- = P$. Then*

$$\gamma_k = (x_+ + x_-)^2.$$

The opposite cross-term

$$\gamma_k^- := A_k^+ + A_k^- - 2P = 2(\alpha_k - P) = \gamma_k - 4P$$

satisfies, in K^/K^{*2} ,*

$$[\gamma_k^-] = [u][\gamma_k]. \quad (9)$$

Thus the two sum/difference choices generate no square class beyond $[u]$ and $[\gamma_k]$.

Proof. The first assertion follows from

$$(x_+ + x_-)^2 = A_k^+ + A_k^- + 2P = 2(\alpha_k + P).$$

Equation (8) reads $\gamma_k \gamma_k^- = 4u\beta_k^2$. Since $4\beta_k^2$ is a square, taking square classes gives (9) wherever the displayed elements are nonzero. Since γ_k is a unit at the even contact constructed below, it is not the zero element of K ; the same identity then shows that the generic square-class statement is well defined. □

Remark 3.4. For $k = 4$, equations (3)–(7) give

$$\alpha_4 = e_4(w) + ue_2(w) + u^2, \quad \beta_4 = e_3(w) + ue_1(w), \quad \gamma_4 = 2(e_4(w) + ue_2(w) + u^2 + P),$$

which is exactly the globally normalized static R9 radicand. The shorter expression $2(P + e_4(w))$ is only its specialization at $u = 0$.

4 Signed-contact divisors

For a sign vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_k) \in \{\pm 1\}^k$, define

$$\ell_\varepsilon = 4M - \sum_{i=1}^k \varepsilon_i N_i, \quad s_\varepsilon = \prod_{i=1}^k \varepsilon_i. \quad (10)$$

The base divisor D_ε is the generic point of $\ell_\varepsilon = 0$, localized away from

$$P \prod_{\eta \neq \varepsilon} \ell_\eta \left(\sum_{i=1}^k \frac{\varepsilon_i}{N_i} \right) = 0. \quad (11)$$

The distinct linear forms ℓ_η define distinct divisors. The final factor in (11) is not the zero rational function: after multiplication by P it is a sum of distinct monomials $\varepsilon_i \prod_{j \neq i} N_j$. Hence this localization is nonempty for every sign vector.

Lemma 4.1 (Simple signed contact). *Above D_ε there is a unique mass sheet on which*

$$u = 0, \quad w_i = \varepsilon_i N_i. \quad (12)$$

On that sheet u is a uniformizer. Equivalently, its valuation satisfies $v_\varepsilon(u) = 1$.

Proof. On the branch through (12), the binomial expansion gives

$$w_i = \varepsilon_i N_i \left(1 + \frac{u}{N_i^2}\right)^{1/2} = \varepsilon_i N_i + \frac{\varepsilon_i}{2N_i} u + O(u^2).$$

Substitution in $4M = \sum_i w_i$ yields

$$\ell_\varepsilon = \frac{u}{2} \sum_{i=1}^k \frac{\varepsilon_i}{N_i} + O(u^2). \quad (13)$$

The coefficient of u is a unit by (11); hence the local implicit-function theorem, or equivalently formal inversion of (13), shows that u and ℓ_ε have the same order. Thus $v_\varepsilon(u) = 1$.

At $u = 0$, every signed radical branch has mass value $\sum_i \eta_i N_i$. On D_ε , the other values differ by the units ℓ_η . Therefore no other mass sheet has $u = 0$ at the generic point of D_ε , proving uniqueness. $\square$

Lemma 4.2 (Contact values of α_k and β_k). *At the contact (12),*

$$\alpha_k(0) = s_\varepsilon P, \quad \beta_k(0) = s_\varepsilon P \sum_{i=1}^k \frac{\varepsilon_i}{N_i}. \quad (14)$$

In particular, β_k is a unit in the localization (11).

Proof. Every term in (3) except $e_k(w)$ is divisible by u , and every term in (4) except $e_{k-1}(w)$ is divisible by u . Hence

$$\alpha_k(0) = \prod_i \varepsilon_i N_i = s_\varepsilon P.$$

Moreover,

$$\begin{aligned} \beta_k(0) &= e_{k-1}(\varepsilon_1 N_1, \dots, \varepsilon_k N_k) \\ &= \sum_{i=1}^k \prod_{j \neq i} \varepsilon_j N_j = s_\varepsilon P \sum_{i=1}^k \frac{\varepsilon_i}{N_i}. \end{aligned}$$

The claimed unit statement follows from (11). $\square$

Proposition 4.3 (The two universal parity rows). *At an even signed contact, $s_\varepsilon = +1$, one has*

$$(v_\varepsilon(u), v_\varepsilon(\gamma_k)) \equiv (1, 0) \pmod{2}.$$

At an odd signed contact, $s_\varepsilon = -1$, one has

$$(v_\varepsilon(u), v_\varepsilon(\gamma_k)) \equiv (1, 1) \pmod{2}.$$

Thus the sheet-level valuation matrix is

$$B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathbf{F}_2). \quad (15)$$

Proof. Lemma 4.1 gives $v_\varepsilon(u) = 1$ in both cases. If $s_\varepsilon = +1$, then Lemma 4.2 gives

$$\gamma_k(0) = 2(P + P) = 4P,$$

which is a unit; hence $v_\varepsilon(\gamma_k) = 0$.

If $s_\varepsilon = -1$, then $\gamma_k(0) = 0$, while $\gamma_k(0) - 4P = -4P$ is a unit. Taking valuations in (8), and using that β_k is a unit, gives

$$v_\varepsilon(\gamma_k) = v_\varepsilon(u) + 2v_\varepsilon(\beta_k) = 1.$$

These are the two rows of (15); its determinant is one in $\mathbf{F}_2$. $\square$

5 Off-sheet parity and orbit saturation

Let E/F be the normal closure of K/F and let

$$I = \text{Hom}_F(K, E), \quad |I| = \delta_k.$$

For $i \in I$, write

$$u_i = i(u), \quad \gamma_i = i(\gamma_k), \quad \beta_i = i(\beta_k).$$

Choose a prime of E above the even or odd divisor D_ε and above the distinguished contact branch of Lemma 4.1. The next lemma is the point that prevents hidden conjugate entries from corrupting the local matrix.

Lemma 5.1 (Off-sheet entries are even). *At such a prime, every off-sheet axial element u_j is a unit. Every off-sheet entropy element γ_j has even valuation. Hence all off-sheet entries in the valuation-parity row are zero.*

Proof. The unique-contact statement in Lemma 4.1 says that only the distinguished mass root vanishes at the generic point of D_ε . Therefore $v(u_j) = 0$ for every off-sheet index j .

All $u_j, w_{r,j}$ and γ_j are regular after the localization (11): the mass polynomial has nonzero leading coefficient there, each $w_{r,j}$ is integral over the corresponding local ring, and γ_j is polynomial in these elements. Thus $v(\gamma_j) \geq 0$. If $v(\gamma_j) = 0$, there is nothing to prove. If it is positive, then P being a unit implies that $\gamma_j - 4P$ is a unit. Applying the conjugate of (8) gives

$$v(\gamma_j) = v(u_j) + 2v(\beta_j) = 2v(\beta_j),$$

which is even. $\square$

Proposition 5.2 (Full conjugate parity matrix). *There exist $2\delta_k$ divisorial valuations of E whose parity matrix on the ordered conjugate radicands*

$$(u_i)_{i \in I} \mid (\gamma_i)_{i \in I}$$

is, after ordering rows and columns,

$$\mathcal{V} = B \otimes I_{\delta_k} = \begin{pmatrix} I_{\delta_k} & 0 \\ I_{\delta_k} & I_{\delta_k} \end{pmatrix}. \quad (16)$$

In particular,

$$\text{rank}_{\mathbf{F}_2}(\mathcal{V}) = 2\delta_k.$$

Proof. Choose one even sign vector and one odd sign vector. Proposition 4.3 and Lemma 5.1 produce the two rows of B , supported on one distinguished embedding. By Theorem 2.1, the group $\text{Gal}(E/F) = S_{\delta_k}$ acts transitively on I . Acting on the selected primes transports each row to every embedding without mixing the two channel coordinates. The resulting matrix is exactly (16). Since B and I_{δ_k} are invertible over $\mathbf{F}_2$, so is their Kronecker product. $\square$

6 The all- k Kummer closure

Define the decorated sheet field

$$H_k = K(\sqrt{u}, \sqrt{\gamma_k}) \quad (17)$$

and its conjugate radical composite

$$L_k = E(\sqrt{u_i}, \sqrt{\gamma_i} : i \in I). \quad (18)$$

The field L_k is the normal closure of H_k/F : every F -embedding sends each displayed square root to a signed square root of another displayed radicand, while L_k contains all such conjugates.

Lemma 6.1 (Valuation rank detects all square classes). *The $2\delta_k$ classes*

$$[u_i], [\gamma_i] \in E^*/E^{*2} \quad (i \in I)$$

are linearly independent over $\mathbf{F}_2$.

Proof. Suppose a product

$$\prod_{i \in I} u_i^{a_i} \gamma_i^{b_i}$$

is a square in E , with $a_i, b_i \in \mathbf{F}_2$. Evaluating its valuation modulo two at each of the $2\delta_k$ primes in Proposition 5.2 says

$$\mathcal{V}(a_i, b_i)_{i \in I}^\top = 0.$$

The matrix $\mathcal{V}$ is invertible, so every a_i and b_i is zero. $\square$

Theorem 6.2 (All- k two-radical Kummer closure). *For every $k \geq 3$, on the generic locus described above,*

$$[H_k : K] = 4, \quad [H_k : F] = 4\delta_k, \quad (19)$$

$$\text{Gal}(L_k/E) \cong (C_2^2)^{\delta_k}, \quad \text{Gal}(L_k/F) \cong C_2^2 \text{ wr } S_{\delta_k}. \quad (20)$$

Consequently,

$$|\text{Gal}(L_k/F)| = 2^{2\delta_k} \delta_k!. \quad (21)$$

Equivalently, the conjugate Kummer relation module is zero.

Proof. Lemma 6.1 proves that the span in E^*/E^{*2} of the $2\delta_k$ conjugate radicands has full dimension $2\delta_k$. Kummer theory in characteristic different from two therefore gives

$$\text{Gal}(L_k/E) \cong (C_2)^{2\delta_k} \cong (C_2^2)^{\delta_k}.$$

In particular, the two classes $[u]$ and $[\gamma_k]$ are independent in K^*/K^{*2} , proving $[H_k : K] = 4$ and hence (19).

Restriction to E gives an exact sequence

$$1 \longrightarrow (C_2^2)^{\delta_k} \longrightarrow \text{Gal}(L_k/F) \longrightarrow S_{\delta_k} \longrightarrow 1.$$

Every automorphism permutes the embedding labels and may change the signs of the two square roots above each label. Thus $\text{Gal}(L_k/F)$ embeds in the permutational wreath product

$$(C_2^2)^{\delta_k} \rtimes S_{\delta_k}.$$

Its kernel is the full base group $(C_2^2)^{\delta_k}$ and its projection to S_{δ_k} is surjective. A subgroup of a semidirect product that contains the full normal factor and surjects onto the quotient is the whole semidirect product. This proves (20); taking orders gives (21). The relation module is the kernel of the map from the free $2\delta_k$ -dimensional permutation module to the conjugate square-class span, and Lemma 6.1 shows that kernel is zero. $\square$

Corollary 6.3 (No additional cross-term channel). *Adjoining $\sqrt{\gamma_k^-}$ to H_k does not enlarge it:*

$$K(\sqrt{u}, \sqrt{\gamma_k}, \sqrt{\gamma_k^-}) = K(\sqrt{u}, \sqrt{\gamma_k}).$$

Proof. Equation (8) gives

$$\sqrt{\gamma_k^-} = \frac{2\beta_k \sqrt{u}}{\sqrt{\gamma_k}}$$

after a compatible choice of signs. □

7 Low- k specializations

The first four nontrivial members are shown in Table 1. The $k = 3, 5, 6$ contact calculations are exact checks of the uniform proof; the $k = 4$ row recovers the established static horizon crown.

Table 1: First all- k two-channel closures.

k	δ_k	Kummer rank	$[H_k : F]$	Normal-closure group
3	4	8	16	C_2^2 wr S_4
4	5	10	20	C_2^2 wr S_5
5	16	32	64	C_2^2 wr S_{16}
6	22	44	88	C_2^2 wr S_{22}

For reference, the explicit symmetric expressions are

$$\begin{aligned}
\alpha_3 &= e_3 + ue_1, & \beta_3 &= e_2 + u, \\
\alpha_4 &= e_4 + ue_2 + u^2, & \beta_4 &= e_3 + ue_1, \\
\alpha_5 &= e_5 + ue_3 + u^2e_1, & \beta_5 &= e_4 + ue_2 + u^2, \\
\alpha_6 &= e_6 + ue_4 + u^2e_2 + u^3, \\
\beta_6 &= e_5 + ue_3 + u^2e_1.
\end{aligned}$$

The alternating pattern is not empirical: it is exactly the parity decomposition proved in Lemma 3.1.

8 Exceptional loci and theorem boundary

The theorem is generic, and its proof identifies the excluded set rather than hiding it in a single unspecified discriminant. The argument localizes away from the following divisors and their higher intersections:

1. $P = 0$, where one or more charge parameters vanish and the norm square P^2 degenerates;
2. $N_i^2 = N_j^2$, where the multiquadratic degree or normalized sheet structure can drop;
3. the reciprocal sub-balance loci $\sum_i \varepsilon_i / N_i = 0$, where the chosen signed contact ceases to be simple or β_k ceases to be a unit;
4. intersections $\ell_\varepsilon = \ell_\eta = 0$ of distinct signed walls, where more than one contact root can occur simultaneously;
5. the base discriminant and leading-coefficient loci of the mass polynomial; and
6. characteristic two, where valuation parity and ordinary quadratic Kummer theory require replacement.

Each exclusion is proper. In particular, the reciprocal sub-balance function is not identically zero, and the signed wall forms are pairwise distinct. Thus the open locus used by the proof is nonempty. Because the base group and Kummer rank are generic open conditions, the intersection of their nonempty open loci is again nonempty in the irreducible parameter space.

For even k , balanced sign classes alter the pole count at infinity and are responsible for the binomial correction in δ_k . They do not affect the finite signed-contact calculation: the local expansion (13), the matrix B , and the off-sheet identity remain unchanged.

9 Consequences

The theorem closes the static all- k Kummer relation-module problem for the canonical norm-derived two-channel cover. Once the symmetric base group is known, no new large Galois computation is required: the full result is forced by two sheet-local contact types and the global identity (8).

The formula also explains why the $k = 4$ calculation was reusable. The published relation

$$[\gamma_4 - 4P] = [u][\gamma_4]$$

is not a special multiplicative coincidence. It is the $k = 4$ member of

$$[\gamma_k - 4P] = [u][\gamma_k]$$

for every k . What is special at four charges is the established physical entropy interpretation, not the underlying Kummer algebra.

The same architecture can be applied to further decorated channels. For any additional canonical radicand, the remaining work is sharply constrained: identify its divisor types, prove regularity, compute the new sheet-level parity rows, and test whether the enlarged matrix remains invertible. The base permutation group and the two channels proved here need not be rebuilt.

Reproducibility note

The exact $k = 3$ and $k = 6$ symbolic identities, contact mass norms, simple-root checks and parity matrices are recorded in the machine-verifiable evidence package [5]. The $k = 5$ specialization was independently replayed with the same exact builder, giving degree 16, contact rows (1, 0) and (1, 1), and Kummer rank 32. These calculations are checks of the theorem, not premises of its proof.

References

- [1] W. Lloyd, *Generic Symmetric Monodromy of Weighted Multiquadratic Sums: A Complete All- k Theorem with a Kummer–Wreath Lifting Criterion*, completed theorem-and-proof paper, v1.0, 16 July 2026.
- [2] W. Lloyd, *Audit Report—R6 + R7 Generic Birationality and Reflection Descent*, proof audit, 2026.
- [3] W. Lloyd, *Conjugate Kummer Modules and Wreath-Product Closures*, general theorem and proof, 10 July 2026.
- [4] W. Lloyd, *Galois Theory of the Horizon Cover: Abel–Ruffini Obstruction, Kummer Lifts, and Wreath-Product Monodromy at Four Charges*, technical preprint, v1.0, 11 July 2026.
- [5] W. Lloyd and OpenAI Codex, *The $k = 3/k = 6$ Radicand Kill Test*, exact machine-verifiable evidence report, v1.0, 18 July 2026.